

Notes: Borjas & Doran (QJE, 2012)

The Collapse of the Soviet Union and the Productivity of American Mathematicians

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1 Big picture

- **Question.** What happens to incumbent knowledge producers when a sudden inflow of highly skilled researchers enters their field? Does the inflow create positive knowledge spillovers, or does it crowd out incumbents through competition for jobs, journal space, citations, and attention?
- **Natural experiment.** The collapse of the Soviet Union suddenly integrated Soviet mathematicians into global mathematics after decades of relative isolation. This shock created variation across mathematical subfields because Soviet and American mathematicians had specialized in different areas before 1992.
- **Treatment intensity.** An American mathematician is more exposed if his or her pre-1990 publication fields overlap more with Soviet pre-collapse research fields.
- **Main result.** American mathematicians with high overlap with Soviet research experienced lower publication and citation output after 1992 than otherwise comparable low-overlap American mathematicians.
- **Extensive-margin effects.** Highly exposed mathematicians were more likely to stop active publishing, more likely to move institutions, and conditional on moving tended to move to lower quality institutions.
- **Top-tail effects.** The probability of producing very highly cited “home run” papers also fell for highly exposed American mathematicians.
- **Student channel.** Students of Soviet emigre advisers had higher productivity, but students of American advisers whose research was Soviet-like had lower productivity after the shock; the aggregate student result reflects both positive mentoring and negative crowd-out channels.
- **Aggregate effect.** Soviet emigres filled in much of the decline in output among incumbent Americans, but the total mathematics “pie” did not appear to expand substantially.
- **International evidence.** Countries that received more Soviet emigres and were more connected to the global mathematics journals market experienced larger productivity losses among their preexisting Soviet-overlap mathematicians.
- **Economic takeaway.** Even in a knowledge market, where ideas are nonrival in principle, scarce jobs, publication slots, audience attention, and frontier researchers’ time can make an inflow of talent crowd out incumbents.

1.1 Incremental contribution of the paper

- **A rare field-level shock to knowledge production.** The paper uses a historically large and plausibly unanticipated shock to the supply of top-tier researchers. This is closer to an economy-wide knowledge-production experiment than the smaller superstar-death or mentor-loss settings common in the literature.
- **Direct measurement of individual scientific production.** It builds a data set from MathSciNet, AMS, ISI, and the Mathematics Genealogy Project that follows individual mathematicians’ publications, citations, fields, institutions, and students over decades.
- **Separates bodies from ideas.** The design distinguishes local labor-market competition from global journal/idea-market competition. The international analysis shows that both the physical arrival of Soviet mathematicians and integration into a global publication market matter.
- **Introduces a quantitative exposure measure.** The overlap indices transform historical field specialization into continuous treatment intensity, allowing a difference-in-differences style comparison within the preexisting American mathematics community.
- **Moves beyond average output.** The paper studies mean productivity, home-run papers, retirement from publishing, institutional mobility, student productivity, aggregate output accounting, and international heterogeneity.
- **Challenges a simple spillover view.** The central contribution is not that knowledge spillovers do not exist, but that positive idea spillovers can be offset or dominated by scarce resources in the markets where knowledge producers compete.

2 Data construction and measurement

2.1 Historical shock and population of interest

The paper studies the post-1992 opening of Soviet mathematics to the United States and the world. Before the collapse, Soviet mathematics was institutionally isolated: international communication and publishing were restricted, collaboration with Western mathematicians was rare, and Soviet mathematical research developed along different subfield lines. After 1992, over 1,000 Soviet mathematicians migrated abroad, with a large group moving to the United States. The paper asks how this affected the preexisting American mathematicians who were already active before the shock.

Interpretation. The key empirical advantage is that the shock was not chosen by American mathematicians and it differentially affected fields. Because Soviet and American subfield specializations were historically different, an incumbent’s pre-1990 field mix predicts exposure to the Soviet inflow.

2.2 AMS/MathSciNet publication data

- The American Mathematical Society provides author-level publication records by field and year.
- Each paper is classified into one of 63 mathematics subject fields.
- The data report institutional affiliation and country of affiliation when available; systematic affiliation data begin around 1984.
- The author-level panel covers long careers and allows the authors to construct annual publications, citations, field shares, and institutional movements.

Interpretation. This data source lets the paper observe the production of codified mathematical knowledge at the individual-year level and at the field-year level.

2.3 Citation measures

The paper uses citation data from two sources:

- AMS citation data, which are incomplete but tied closely to the MathSciNet universe.
- ISI citation data, which provide a more standard citation measure for matched papers.

Interpretation. Publications measure output quantity; citations proxy for audience attention and quality. Using both helps show that the result is not only a decline in low-value papers.

2.4 Defining American mathematicians and Soviet emigres

- A preexisting American mathematician is typically a mathematician predominantly affiliated with a U.S. institution before 1990 and active before the Soviet shock.
- A Soviet emigre is a mathematician who was predominantly affiliated with a Soviet institution before 1990 and later had a modal affiliation outside the former Soviet Union.
- The analysis distinguishes Soviet emigres to the United States, Soviet emigres to other countries, and Soviet mathematicians who remained in the former Soviet area.

Interpretation. The classification separates the incumbent workforce from the incoming shock and makes it possible to compare the productivity of movers, stayers, and incumbent Americans.

2.5 Research fields and exposure to the Soviet shock

Let a_{ij} be the share of mathematician i 's pre-1990 papers in field j , and let s_j be the share of Soviet papers in field j before the collapse. The paper uses three overlap indices.

Correlation coefficient. The first index is the correlation between the American mathematician's field-share vector and the Soviet field-share vector.

Index of intensity.

$$I_i = \frac{\sum_j a_{ij} S_j}{\max_j (S_j)},$$

where S_j is the total number of Soviet papers in field j .

Index of similarity.

$$D_i = 1 - \frac{1}{2} \sum_j |a_{ij} - s_j|.$$

Interpretation. The three indices capture related but distinct ideas: correlation captures relative similarity; intensity weights the American mathematician's fields by the absolute number of Soviet papers; similarity measures the overlap in field distributions.

2.6 Output, top-tail output, mobility, and survival

- **Annual productivity:** number of papers and number of citations.
- **Home runs:** papers in the extreme upper tail of the citation distribution, such as above the 95th percentile.
- **Retirement from publishing:** cessation of observed publishing, modeled using survival analysis.
- **Institutional mobility:** movement between pre-1990 and post-1992 modal affiliations.
- **Institutional quality:** based on total mathematical publications by institutions.
- **Student productivity:** annual papers and citations of doctoral students, using Mathematics Genealogy Project advisor-student links.

Interpretation. The outcomes distinguish crowd-out at different margins: average output, elite output, labor-market displacement, exit from publishing, and intergenerational productivity.

2.7 International data

For the international analysis, countries are included if they had at least 500 active mathematicians before the collapse. The paper constructs:

- the number of preexisting mathematicians by country;
- the number of Soviet emigres who moved to each country;
- a quality-adjusted immigrant share based on citations;
- a connectivity index measuring how connected a country is to the global publication market.

Interpretation. The international setting helps separate local job-market crowd-out from global journal-market competition. A country can experience idea-market competition even if few emigres physically move there.

3 Empirical specifications

3.1 Baseline author-year difference-in-differences specification

The main author-year specification is:

$$y_i(t) = \alpha_i + \alpha_t + X_i(t)' \gamma + \theta (Post_t \times Index_i) + \varepsilon_i(t),$$

where $y_i(t)$ is annual papers or citations for mathematician i in year t , α_i are individual fixed effects, α_t are year fixed effects, $X_i(t)$ includes a quartic in experience, $Post_t$ equals one after 1992, and $Index_i$ is one of the overlap indices.

Interpretation. The coefficient θ compares the post-1992 output change of high-overlap mathematicians to the post-1992 output change of low-overlap mathematicians, net of individual fixed differences, aggregate year shocks, and career-cycle controls.

3.2 Author-field-year specification

To rule out arbitrary field-specific trends, the paper also estimates:

$$y_{ij}(t) = \alpha_i + \alpha_j + \alpha_t + (\alpha_j \times \alpha_t) + X_i(t)' \gamma + \theta (Post_t \times Index_i) + \varepsilon_{ij}(t),$$

where j is a mathematics field and $(\alpha_j \times \alpha_t)$ are field-year fixed effects.

Interpretation. This is a demanding specification because it controls for any time-varying shock to a field. The identifying variation comes from differences in individual exposure within the same field-year environment.

3.3 Home-run regressions

The paper estimates Poisson models for the expected number of home-run papers:

$$E(H_{it}|X_{it}) = \exp\{\alpha + \delta Post_t + \beta X_{it}\},$$

estimated separately for low-, middle-, and high-exposure groups.

Interpretation. This asks whether the Soviet shock affected the upper tail of scientific output, not just the average publication count.

3.4 Retirement from publishing

The survival analysis uses a Cox proportional hazard model:

$$\log \lambda_i(Z_i, \tau) = \log \lambda_0(\tau) + Z_i' \phi + \varepsilon_i,$$

where τ is experience and the key covariates include the overlap index and post-1992 exposure.

Interpretation. The hazard model measures whether the Soviet shock pushed high-overlap mathematicians out of active publishing earlier in their careers.

3.5 Institutional mobility and quality

The mobility regressions relate whether a mathematician moved institutions and the quality of the new institution to:

- intellectual exposure: the overlap index;
- geographic exposure: whether the pre-1990 institution hired Soviet emigres;
- controls for individual productivity, experience, and pre-shock institutional quality.

Interpretation. These specifications test whether crowd-out operates through local academic labor markets in addition to the global market for journal space.

3.6 Student productivity

The student regressions relate students' later annual papers and citations to whether their advisor was a Soviet emigre or an American mathematician with high Soviet-style research overlap.

Interpretation. This separates a positive mentoring channel from the negative productivity effect on American advisors whose own research programs were crowded out.

3.7 Aggregate counterfactual and international regressions

For aggregate output, the paper estimates pre-1992 age-output profiles and predicts post-1992 output for different exposure groups. For international heterogeneity, it estimates first-stage productivity impacts by country and then relates those impacts to immigrant shares and journal connectivity:

$$\hat{\theta}_c = \beta_0 + \beta_1 p_c + \beta_2 V_c + u_c,$$

where p_c is the quality-adjusted immigrant share and V_c is the country connectivity index.

Interpretation. The aggregate exercise asks whether Soviet output filled the incumbent output gap. The international exercise tests whether local bodies and global ideas both contribute to productivity crowd-out.

4 Identification strategy

4.1 Core source of identification

The design combines a large historical shock with predetermined exposure. The collapse of the Soviet Union was abrupt and not plausibly caused by trends in U.S. mathematics. Exposure is measured using pre-1990 publication fields, before the shock changed the global market. The core comparison is therefore:

Did American mathematicians whose pre-shock research overlapped more with Soviet mathematics experience a larger post-1992 productivity decline than American mathematicians whose pre-shock research overlapped less?

4.2 Why the shock is plausibly exogenous

- Soviet and American mathematicians had developed separate specializations over decades.
- Pre-1990 collaboration between Soviet and American mathematicians was minimal.
- The timing of the political collapse was widely unexpected.
- The influx was large and sudden relative to the preexisting U.S. mathematical workforce.

Interpretation. The shock is not a policy assigned randomly to fields, but the historical setting makes it difficult to argue that U.S. mathematicians with Soviet-style research anticipated the event and adjusted before 1992.

4.3 Why field overlap identifies exposure

Fields with a high Soviet-to-American pre-shock publication ratio received more new Soviet competition and more Soviet ideas after 1992. An American mathematician working in those fields was more exposed than one working in fields such as statistics or computer science, where the Soviet pre-shock presence was small.

Interpretation. The exposure variable captures both potential spillovers and potential competition. The sign of the coefficient is therefore an empirical statement about the net effect.

4.4 Key controls and fixed effects

- Individual fixed effects absorb fixed differences in ability, productivity, and career orientation.
- Year fixed effects absorb aggregate shocks to mathematics output and publishing norms.
- Experience polynomials control for career life-cycle productivity.
- Field-year fixed effects address the possibility that some subfields had independent post-1992 trends.
- Standard errors are clustered at the individual level.

Interpretation. The strongest specifications ask whether more-exposed individuals changed relative to less-exposed individuals within a field-year environment, rather than whether Soviet-style fields as a whole declined.

4.5 Threats to identification

- **Differential pre-trends.** If high-overlap mathematicians were already declining before 1992, the treatment estimate could overstate the shock. Figure VII helps address this by showing pre-period trends.
- **Field-specific shocks.** A field might decline for reasons unrelated to Soviet entry. Author-field-year specifications with field-year effects are designed to handle this.
- **Measurement of citations.** Citation data differ across AMS and ISI sources. The paper uses multiple citation measures and also examines publications.
- **Selection into emigration.** Soviet emigres were positively selected. This strengthens the competitive shock but means the paper estimates the effect of a selected high-quality inflow rather than a random inflow of average mathematicians.
- **Collaboration benefits.** Some incumbents coauthored with Soviet emigres and likely gained. The paper explicitly tests the coauthor subgroup and the student/adviser channel.
- **General equilibrium.** The estimates capture net effects in a constrained knowledge market. They do not imply that ideas themselves are rival; rather, they show that surrounding resources can be scarce.

4.6 Why the international section strengthens the design

The country-level analysis exploits variation in where Soviet mathematicians moved and how connected countries were to the global publication market. If the U.S. results were only a local labor-market story, countries with few emigres but high journal connectivity should not show similar effects. The evidence that both emigre share and connectivity matter supports the paper's interpretation that the shock operated through both bodies and ideas.

5 Tables and figures

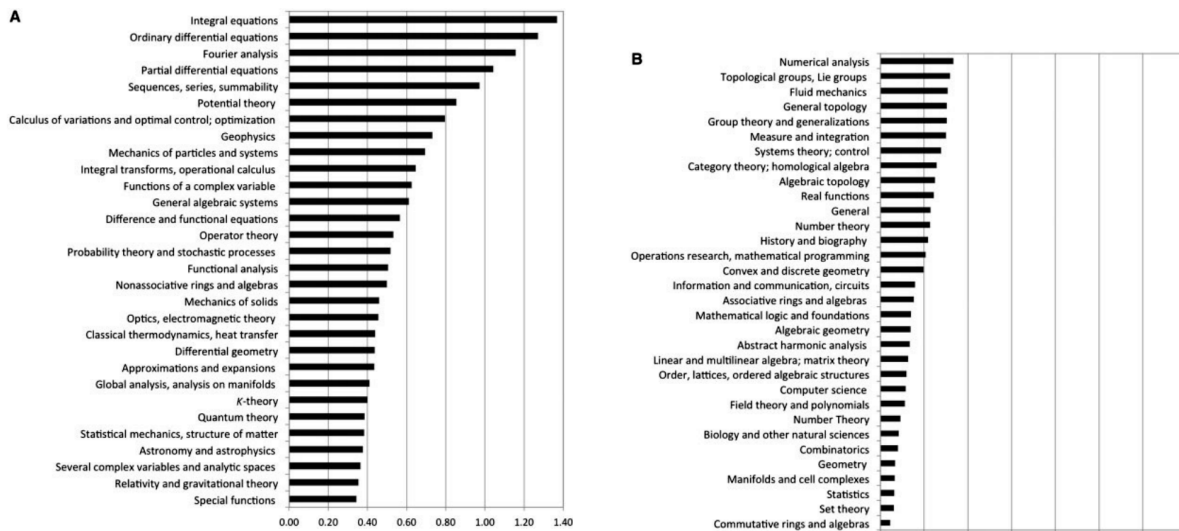
5.1 Figure I: Ratio of Soviet Papers to American Papers, by Field, 1984-1989

Read:

- Panel A shows fields where Soviet publication intensity was high relative to American publication intensity, such as integral equations, ordinary differential equations, Fourier analysis, and partial differential equations.
- Panel B continues the list and shows fields with low Soviet relative presence, including statistics, computer science, combinatorics, and algebraic geometry.
- The dispersion is large: some fields have more Soviet papers per American paper, while others are heavily American-dominated.

Interpretation. Figure I is the foundation of the treatment design. It shows that Soviet and American mathematics had different pre-shock field specializations, making exposure heterogeneous across incumbent American mathematicians.

Economic message. The shock was not a uniform increase in scientific labor. It was concentrated in specific fields, which creates usable variation for identifying productivity effects.



(a) Panel A: Soviet-intensive fields

(b) Panel B: American-intensive fields

5.2 Figure II: Trend in Coauthorship Rate between Soviet and American Mathematicians and Figure III: Employment Trends for New Mathematics Doctorates

Read:

- Figure II shows that Soviet-American coauthorship was almost nonexistent before 1990 and rose sharply only after the Soviet collapse.
- Figure III shows deterioration in the academic employment market for new North American mathematics PhDs around the early 1990s: unemployment rose and the fraction employed at PhD-granting institutions fell.

Interpretation. Together these figures document both the idea channel and the labor-market channel: contact and collaboration increased, but so did job-market pressure.

Economic message. The setting contains both possible positive spillovers and possible crowd-out. The empirical question is which force dominates for exposed incumbents.

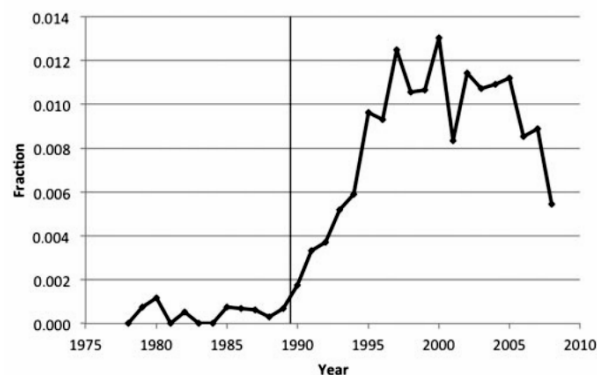
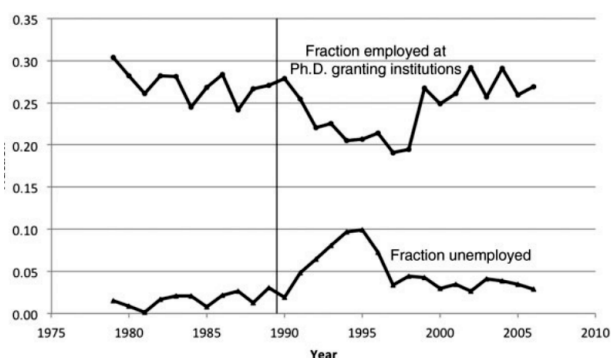


FIGURE II

(a) Figure II: coauthorship rate



(b) Figure III: new doctorate employment

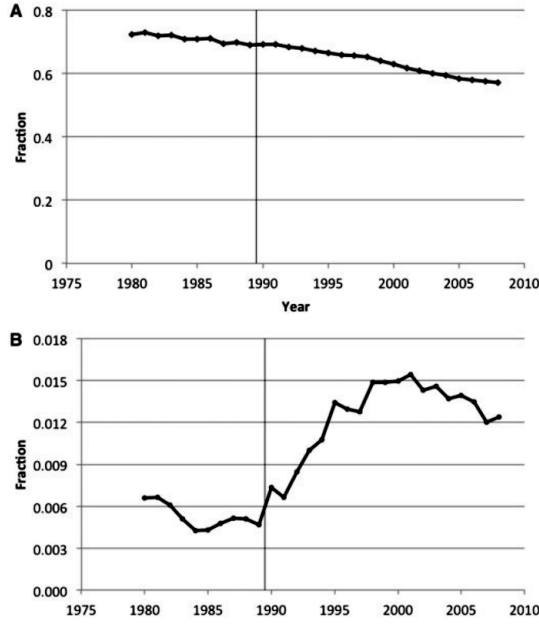
5.3 Figure IV: Citations to American or Soviet Papers by American Mathematicians and Figure V: Frequency Distribution of Output for American Mathematicians

Read:

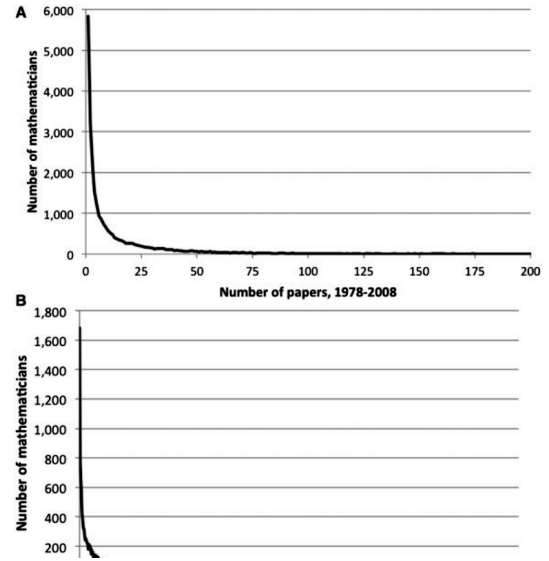
- Figure IV shows a relatively stable or declining share of citations to American papers and a large rise in citations to Soviet papers after 1990.
- Figure V shows that output is extremely skewed: many American mathematicians publish few papers and receive few citations, while a small right tail is very productive.

Interpretation. Figure IV confirms that American mathematicians began using Soviet research more after the collapse. Figure V motivates looking beyond means because the productivity distribution is highly skewed.

Economic message. The shock increased access to Soviet knowledge, but measuring its productivity effect requires examining both average and tail output.



(a) Figure IV: citations to American/Soviet papers



(b) Figure V: output distribution

5.4 Table I: Summary Statistics for Samples of American and Soviet Mathematicians and Figure VI: Fraction of Publications Published by Soviet Emigres, by Type of Field

Read:

- Table I shows that Soviet emigres were positively selected: before 1992, future emigres had more papers and citations than other Soviets, and after 1992 they were far more productive than the average American mathematician.
- Soviet emigres to the United States also had unusually broad field coverage, with a much higher median number of fields.
- Figure VI shows that Soviet emigre output rose sharply in Soviet-style fields but much less in U.S.-style fields.

Interpretation. The incoming group was not just large; it was highly productive and concentrated in particular research areas. This makes the supply shock economically meaningful.

Economic message. A shock from highly selected researchers is likely to affect both competition for jobs and competition for attention in journals.

TABLE I
SUMMARY STATISTICS FOR SAMPLES OF AMERICAN AND SOVIET MATHEMATICIANS

Variable	Group of mathematicians			
	Americans	Soviet émigrés to U.S.	Soviet émigrés elsewhere	All other Soviets
Number of mathematicians	29,392	336	715	11,173
Papers published, 1978–1991				
Mean papers per mathematician	6.7	17.8	14.6	8.1
Median papers	3.0	13.0	10.0	5.0
Maximum number of papers	232.0	104.0	152.0	180.0
Papers published, 1992–2008				
Mean papers per mathematician	6.8	27.2	28.8	7.6
Median papers	1.0	21.0	22.0	1.0
Maximum number of papers	768.0	128.0	317.0	311.0
Citations, AMS, 1978–1991				
Mean citations per mathematician	29.1	74.6	32.8	8.6
Median citations	1.0	10.0	6.0	0.0
Maximum number of citations	5550.0	1276.0	1441.0	2928.0
Citations, AMS, 1992–2008				
Mean citations per mathematician	33.6	177.4	110.3	13.4
Median citations	0.0	62.0	37.0	0.0
Maximum number of citations	3404.0	1709.0	1988.0	1287.0
Citations, ISI, 1978–1991				
Mean citations per mathematician	110.2	185.1	79.8	25.3
Median citations	20.0	25.5	11.0	3.0
Maximum number of citations	20,274.0	7232.0	3040.0	3054.0
Citations, ISI, 1992–2008				
Mean citations per mathematician	52.1	209.0	156.2	27.3
Median citations	0.0	88.5	60.0	0.0
Maximum number of citations	11,688.0	3371.0	4442.0	1258.0
Median number of fields	2.0	5.5	5.0	2.0
Percent first published after 1980	45.2	40.5	46.7	48.8

Notes: The sample consists of "active" mathematicians who published at least one paper between 1970

(a) Table I: summary statistics

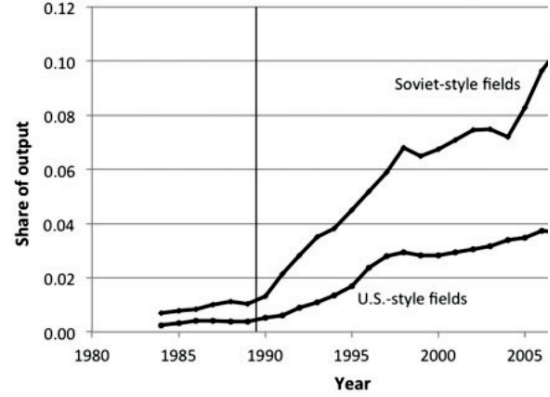


FIGURE VI

Fraction of Publications Published by Soviet Emigrés, by 1
The U.S.-style fields consist of the bottom 10 fields in Fig
(b) Figure VI: emigre output share

5.5 Table II: American Mathematicians with the Largest Number of Publications, 1960-1989

Read:

- Table II lists highly productive American mathematicians and reports their overlap with Soviet mathematics using the correlation coefficient, intensity index, and similarity index.
- Even among superstars, overlap varies substantially: some highly productive mathematicians worked in Soviet-like fields, others did not.

Interpretation. This table helps validate the exposure measure. It is not merely a proxy for overall productivity because top American mathematicians can have high or low Soviet overlap.

Economic message. The treatment intensity is about research direction, not simply talent.

TABLE II
AMERICAN MATHEMATICIANS WITH THE LARGEST NUMBER OF PUBLICATIONS, 1960–1989

Mathematician	Number of papers	Correlation coefficient	Index of intensity	Index of similarity
Frank Harary	416	−0.039	0.101	0.096
Pranab Kumar Sen	318	0.035	0.199	0.090
Richard Ernest Bellman	317	0.420	0.339	0.410
Ciprian Foias	299	0.295	0.414	0.291
Avner Friedman	290	0.629	0.633	0.411
Robert E. Kalaba	241	0.272	0.306	0.308
Peter C. Fishburn	232	0.157	0.302	0.126
Bang-Yen Chen	220	0.009	0.164	0.070
Barry Simon	217	0.537	0.580	0.325
V. Lakshmikantham	215	0.467	0.633	0.310

Notes: The correlation coefficient, index of intensity, and index of similarity measure each mathematician's research overlap with the Soviet research program for papers published between 1960 and 1989. See the text for the definition of the various indices.

Table II: overlap indices for prolific American mathematicians

5.6 Table III: Impact of Soviet Supply Shock on American Mathematicians and Figure VII: Impact of Index of Similarity on Output of American Mathematicians

Read:

- Table III reports negative post-1992 interaction coefficients for all three overlap measures and for both papers and citations.
- The main author-year estimates imply that moving from no overlap to high overlap substantially reduces post-1992 publication output.
- The author-field-year estimates remain negative after controlling for field-year shocks.
- Figure VII shows that high-exposure mathematicians had different post-1992 output trajectories: output trends turn down sharply after 1990 relative to the low-exposure group.

Interpretation. Table III is the core empirical result. The same negative pattern appears across alternative exposure measures, alternative samples, short-run and long-run windows, and field-year specifications.

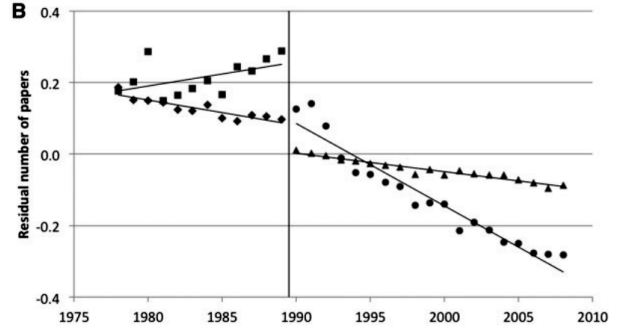
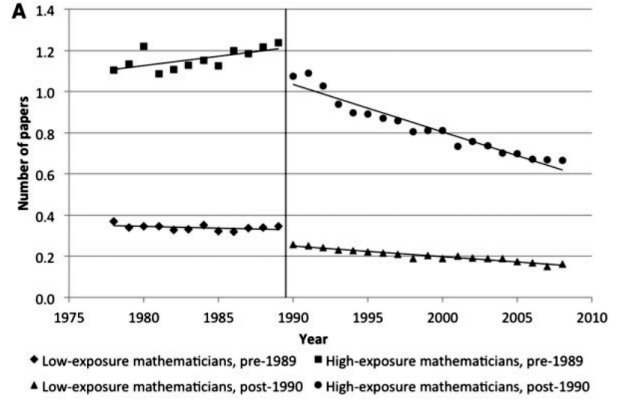
Economic message. The results suggest that the arrival of Soviet mathematicians reduced the productivity of incumbent Americans working in nearby research areas.

TABLE III
IMPACT OF SOVIET SUPPLY SHOCK ON AMERICAN MATHEMATICIANS

Specification/regressor	Mathematicians predominantly in U.S.		Mathematicians always in U.S.	
	Number of papers	Number of citations	Number of papers	Number of citations
A. Author-year regressions				
Correlation coefficient	-0.133 (0.036)	-19.577 (1.576)	-0.116 (0.034)	-16.298 (1.540)
Index of intensity	-0.047 (0.028)	-14.845 (1.293)	-0.042 (0.027)	-12.293 (1.261)
Index of similarity	-1.523 (0.113)	-69.155 (4.645)	-1.419 (0.108)	-58.494 (4.655)
B. Author-year regressions, short run				
Correlation coefficient	-0.102 (0.032)	-14.214 (1.783)	-0.085 (0.030)	-11.404 (1.410)
Index of intensity	-0.045 (0.023)	-10.944 (1.221)	-0.039 (0.022)	-8.830 (1.181)
Index of similarity	-1.056 (0.111)	-48.547 (4.232)	-0.985 (0.108)	-39.054 (4.117)
C. Author-year regressions, long run				
Correlation coefficient	-0.122 (0.049)	-25.219 (2.037)	-0.108 (0.046)	-21.095 (2.019)
Index of intensity	-0.019 (0.039)	-19.179 (1.687)	-0.015 (0.037)	-15.889 (1.666)
Index of similarity	-1.930 (0.150)	-91.211 (5.961)	-1.802 (0.145)	-77.930 (6.055)
D. Author-field-year regressions				
Correlation coefficient	-0.0021 (0.0006)	-0.3048 (0.0249)	-0.0020 (0.0005)	-0.2578 (0.0244)
Index of intensity	-0.0007 (0.0004)	-0.2378 (0.0206)	-0.0007 (0.0004)	-0.2005 (0.0202)
Index of similarity	-0.0238 (0.0017)	-1.0248 (0.0732)	-0.0240 (0.0016)	-0.8696 (0.0732)

Notes: Standard errors are reported in parentheses and clustered at the individual level. The table reports the coefficient of the interaction between the overlap index and a post-1992 dummy variable. Each observation in Panels A, B, and C represents a unique author-year permutation; an observation in Panel D represents a unique author-field-year permutation. The regressions use the entire sample period 1978–2008 in Panels A and B, the 1978–1999 period in the short run, and the 1978–1991 and 2000–2008 periods in the long run. The numbers of papers (citations) regressions in the predominantly American sample

(a) Table III: productivity regressions



(b) Figure VII: output by exposure

5.7 Table IV: Impact of Soviet Supply Shock on Probability of “Home Run” Papers

Read:

- The table estimates changes in the expected number of papers above high citation thresholds.
- Highly exposed American mathematicians experienced a large decline in home-run production after 1992.
- Low-exposure mathematicians show little or no comparable decline.
- The coauthorship columns suggest that the average negative effect is not simply offset by collaboration with Soviet emigres.

Interpretation. The Soviet shock reduced not only the quantity of output but also the likelihood of exceptional, high-impact papers among highly exposed incumbents.

Economic message. Competition for frontier attention and journal space can affect the upper tail of knowledge production.

TABLE IV
IMPACT OF SOVIET SUPPLY SHOCK ON NUMBER OF “HOME RUNS” (CONDITIONAL FIXED
EFFECTS POISSON MODEL)

Sample	Number of papers with citations above x th percentile:			By coauthorship status (95th percentile)	
	90th	95th	99th	Coauthored with émigré	Did not coauthor
Correlation coefficient in:					
Bottom quartile	0.058 (0.066)	-0.055 (0.097)	0.211 (0.237)	0.005 (0.279)	-0.055 (0.103)
Middle 50%	0.012 (0.032)	0.029 (0.044)	0.120 (0.090)	0.058 (0.144)	0.021 (0.046)
Top quartile	-0.204 (0.032)	-0.160 (0.043)	-0.121 (0.087)	0.014 (0.091)	-0.215 (0.049)
Index of intensity in:					
Bottom quartile	0.089 (0.068)	-0.025 (0.101)	0.127 (0.260)	0.022 (0.286)	-0.025 (0.108)
Middle 50%	-0.043 (0.031)	-0.025 (0.042)	0.112 (0.086)	0.040 (0.131)	-0.036 (0.044)
Top quartile	-0.166 (0.034)	-0.123 (0.045)	-0.115 (0.090)	0.019 (0.095)	-0.169 (0.051)
Index of similarity in:					
Bottom quartile	0.331 (0.069)	0.366 (0.095)	0.402 (0.198)	-0.190 (0.507)	0.393 (0.096)
Middle 50%	0.011 (0.037)	-0.020 (0.050)	-0.004 (0.106)	-0.096 (0.162)	-0.024 (0.053)
Top quartile	-0.158 (0.029)	-0.122 (0.039)	0.016 (0.079)	0.072 (0.085)	-0.180 (0.044)

Notes: Standard errors reported in parentheses. The sample consists of predominantly American mathematicians. An observation consists of an author-year pair for each year between 1978 and 2005. The dependent variable is the number of “home runs,” defined as the number of papers written in that year that have received a sufficiently high number of citations to place them in the top 90th, 95th, or 99th percentile for papers published that year. The regression includes a post-1992 dummy variable, years of experience (introduced as a quartic), and individual fixed effects. The coefficient reported in the table is that of the post-1992 dummy variable. Each coefficient in the table is drawn from a separate Poisson regression estimated in the respective sample. The regressions in the last two columns reestimate the 95th percentile regressions separately on the samples of American mathematicians who either coauthored or did not coauthor with a Soviet émigré at any point after 1992. The number of observations for the

Table IV: home-run papers

5.8 Table V: Impact of Soviet Supply Shock on Probability of Retirement from Publishing and Figure VIII: Survival Functions for American Mathematicians, by Level of Exposure

Read:

- Table V reports positive hazard coefficients: higher overlap predicts a greater probability of exit from active publishing after the Soviet influx.
- The effect is strongest for mathematicians with less than 10 years of experience.
- Figure VIII shows survival curves falling faster after the Soviet influx, especially for the high-exposure group.

Interpretation. The shock affected the extensive margin of scientific careers. Younger and more exposed mathematicians were more likely to leave active publishing.

Economic message. Talent inflows can reallocate scientific careers by pushing marginal incumbents out of the active publication market.

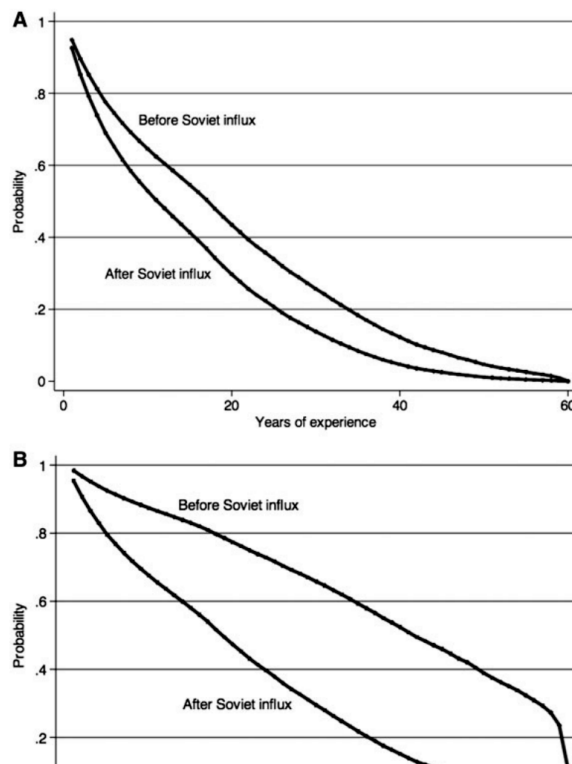
TABLE V
IMPACT OF SOVIET SUPPLY SHOCK ON PROBABILITY OF “RETIREMENT” FROM PUBLISH
(COX PROPORTIONAL HAZARD MODELS)

Sample	Measure of overlap		
	Correlation coefficient	Index of intensity	Index similar
All preexisting mathematicians	0.410 (0.090)	0.230 (0.084)	5.57 (0.29)
Less than 10 years of experience	1.099 (0.229)	0.653 (0.176)	10.34 (0.96)
10–19 years of experience	0.166 (0.192)	0.299 (0.175)	0.23 (0.64)
At least 20 years of experience	0.099 (0.181)	0.101 (0.183)	1.43 (0.49)

Notes: Standard errors are reported in parentheses and are clustered at the individual level. Sample consists of predominantly American mathematicians. Each observation in the data represent individual-year of experience record, beginning in 1978 until the year of the individual's last observed publication. The regression model includes a post-1992 dummy variable, the mathematician's overlap index, and the interaction between the overlap index and the post-1992 indicator (which is the coefficient reported in the table). The sample sizes (by row) are 446,720 observations, 151,012 observations, 143 observations, and 152,267 observations.

mathematician's overlap index, the post-1992 indicator variable and the interaction between these two variables. The top row of Table V reports the coefficient of this interaction variable as

(a) Table V: publishing retirement hazards



(b) Figure VIII: survival functions

5.9 Table VI: Impact of Soviet Supply Shock on Institutional Mobility

Read:

- Table VI shows that both intellectual exposure and geographic exposure predict institutional movement.
- Mathematicians at institutions that hired Soviet emigres were more likely to move.
- Conditional on moving, exposed mathematicians tended to move to lower quality institutions.

Interpretation. This table provides evidence for a labor-market channel. The shock was not only about citations and journal competition; it changed employment trajectories.

Economic message. Highly skilled immigration can crowd out incumbents through local job markets even in a specialized academic field.

5.10 Table VII: Impact of Soviet Supply Shock on Productivity of Students

Read:

- Students of Soviet emigre advisors produced more papers and citations than students of comparable non-emigre advisors.
- Students of American advisors with Soviet-like research programs had lower productivity after 1992.
- Advisor fixed effects reduce but do not eliminate the negative pattern tied to American advisors' overlap.

TABLE VI
IMPACT OF SOVIET SUPPLY SHOCK ON INSTITUTIONAL MOBILITY

Sample/regressor	Measure of overlap/dependent variable					
	Correlation coefficient		Index of intensity		Index of similarity	
	Moved	Δ Quality	Moved	Δ Quality	Moved	Δ Qua
A. All mathematicians						
Institution hired émigré	0.046 (0.013)	-2.382 (0.122)	0.046 (0.013)	-2.383 (0.122)	0.047 (0.013)	-2.38 (0.12)
Overlap index	0.172 (0.025)	-0.415 (0.308)	0.158 (0.022)	-0.282 (0.252)	0.321 (0.066)	-1.32 (0.96)
B. First published after 1980						
Institution hired émigré	0.098 (0.020)	-2.408 (0.158)	0.098 (0.020)	-2.411 (0.158)	0.099 (0.020)	-2.40 (0.15)
Overlap index	0.216 (0.040)	-0.618 (0.373)	0.157 (0.031)	-0.394 (0.285)	0.676 (0.151)	-2.60 (1.48)
C. First published before 1980						
Institution hired émigré	0.001 (0.015)	-2.274 (0.188)	0.001 (0.015)	-2.277 (0.189)	0.001 (0.015)	-2.27 (0.18)
Overlap index	0.100 (0.029)	0.225 (0.538)	0.099 (0.029)	0.356 (0.540)	0.218 (0.069)	-0.18 (1.34)

Notes: Robust standard errors are reported in parentheses. The sample consists of predominant American mathematicians whose last observed affiliation before 1990 and the modal post-1992 observ were both in the United States. The dependent variable in the "moved" regressions is a dummy vari

Table VI: institutional mobility

TABLE VII
IMPACT OF SOVIET SUPPLY SHOCK ON PRODUCTIVITY OF STUDENTS (DEPENDENT VARIABLE=STUDENT'S ANNUAL NUMBER OF PAPERS OR CITATIONS)

Index/variable	Papers			Citations (ISI)		
	(1)	(2)	(3)	(1)	(2)	(3)
Correlation coefficient						
Soviet advisor	0.225 (0.040)	0.232 (0.041)	—	0.557 (0.211)	0.693 (0.223)	—
Advisor's index $\times T$	-0.425 (0.092)	-0.386 (0.091)	-0.178 (0.106)	-6.936 (1.124)	-6.482 (1.127)	-3.808 (1.445)
Index of intensity						
Soviet advisor	0.256 (0.041)	0.265 (0.042)	—	1.147 (0.225)	1.309 (0.236)	—
Advisor's index $\times T$	-0.508 (0.095)	-0.465 (0.095)	-0.236 (0.106)	-9.036 (1.346)	-8.495 (1.349)	-5.429 (1.576)
Index of similarity						
Soviet advisor	0.295 (0.041)	0.296 (0.042)	—	0.875 (0.223)	0.877 (0.235)	—
Advisor's index $\times T$	-0.408 (0.187)	-0.369 (0.182)	-0.129 (0.242)	-5.529 (1.944)	-5.143 (1.976)	-1.302 (3.073)
Institution fixed effects	No	Yes	Yes	No	Yes	Yes
Advisor's fixed effects	No	No	Yes	No	No	Yes

Notes: Robust standard errors are reported in parentheses. The dependent variable is the average annual number of papers published or (ISI) citations received between the time the student obtained his doctoral degree and 2011. The sample consists of students who received doctoral degrees from American institutions between 1978 and 2008, and whose advisor was either a predominantly American preexisting mathematician or a Soviet émigré. The papers regressions have 20,822 observations, and the citations regressions have 20,743 observations. The "Soviet Advisor" variable is a dummy variable set to unity if the advisor is a Soviet émigré; the "Advisor index" variable is the American's advisor's value of the overlap index (set to 0 if the advisor is a Soviet émigré); and T is a dummy variable set to unity for the post-1992

Table VII: student productivity

Interpretation. The student channel is mixed. Soviet emigres transmitted valuable knowledge to students, but the negative shock to American advisors also affected the next generation.

Economic message. Knowledge spillovers and crowd-out can coexist within the same market and even within the same training pipeline.

5.11 Table VIII: Predicted Annual Output of American Mathematicians

Read:

- The paper predicts post-1992 output using pre-1992 age-output profiles.
- Actual output for the most exposed American mathematicians falls below the predicted path.
- Soviet emigre output fills much of the gap, but total output does not rise dramatically above the counterfactual path.

Interpretation. This table converts individual-level estimates into an aggregate accounting exercise. It asks whether the U.S. mathematics pie became larger or whether Soviet output substituted for incumbent output.

Economic message. The paper's answer is closer to substitution than expansion: the inflow replaced output that would otherwise have been produced by incumbents.

TABLE VIII
PREDICTING ANNUAL OUTPUT OF AMERICAN MATHEMATICIANS

	American mathematicians					
	Most exposed	Middle 50%	Least exposed	All Americans	Soviet émigrés	Total
Number of papers, short-run prediction						
Actual	4015.0	3606.6	1253.3	8874.9	370.6	9244.5
Predicted	5062.0	3519.2	900.6	9481.8	—	9482.4
Difference	−1047.0 (121.2)	87.4 (126.5)	352.7 (72.1)	−606.9 (189.5)	—	−233.5 (189.5)
Long-run prediction:						
Number of papers						
Actual	3350.6	3013.1	1035.3	7398.9	432.5	7830.3
Predicted	3879.9	2910.3	757.4	7547.5	—	7547.5
Difference	−529.3 (76.3)	102.8 (119.0)	277.9 (88.0)	−148.6 (166.5)	—	281.4 (166.5)
Number of citations						
Actual	19,355.5	13,372.8	3939.2	36,667.5	2944.8	39,611.0
Predicted	22,088.5	13,191.7	2914.3	38,194.5	—	38,194.5
Difference	−2733.0 (1083.2)	181.1 (1693.2)	1024.9 (1239.9)	−1527.0 (2361.7)	—	141.5 (2361.7)

Notes: The standard errors reported in parentheses give the forecast error of the predicted number of papers or citations, adjusted for the correlation across observations for the same individual. The sample consists of persons who were always affiliated with an American institution prior to 1990. The predicted number of papers in the short run is based on a regression estimated separately in each of the three samples of mathematicians ranked according to their placement in the distribution of the coefficient of similarity. The unit of observation is an author-year; the regression uses observations between 1970 and 1991; and the regressors include a quartic in experience and individual fixed effects. We then use the regression to predict each mathematician's output in each year between 1992 and 1999, and the table reports the sum of these predictions. The long-run predictions are based on the specification estimated over the entire sample period 1978–2008 and reported in Panel A of Table III (using the index of similarity). This regression is then used to predict a mathematician's output in each year between 1992 and 2008 assuming that no Soviet influx occurred (so the index of similarity plays no differential role between the two groups).

Table VIII: aggregate predicted and actual output

5.12 Table IX: The Diaspora of Soviet Mathematicians after the Collapse of the Soviet Union and Figure IX: International Differences in Productivity Impact of Soviet Influx

Read:

- Table IX lists destination countries, preexisting mathematics communities, numbers of Soviet emigres, quality-adjusted immigrant shares, and connectivity indices.
- The United States received 336 emigres, about 32 percent of the measured diaspora.
- Figure IX shows that countries with larger quality-adjusted immigrant shares and stronger global journal connectivity tended to have more negative productivity impacts.

Interpretation. The international results show that competition was not purely local. Countries connected to the global publication market were exposed to idea-market competition even when emigre inflows varied.

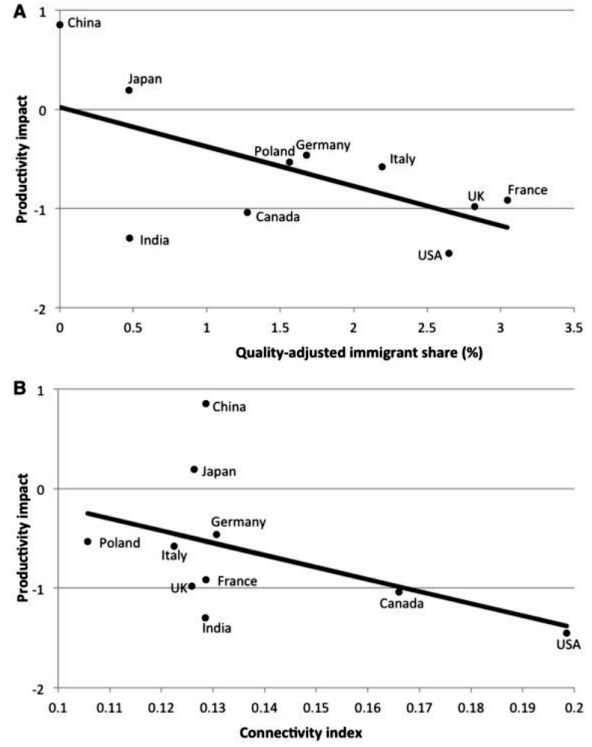
Economic message. The relevant market for knowledge production is partly global; journal space and attention connect researchers across countries.

TABLE IX
THE DIASPORA OF SOVIET MATHEMATICIANS AFTER THE COLLAPSE OF THE SOVIET UNION

	First-stage coefficient	Number of mathematicians	Number of emigrants	Immigrant share (%)	Quality-adjusted share (%)	Connectivity index
ntes	-1.452**	22,449	336	1.50	2.65	0.199
	0.853*	6563	0	0.00	0.00	0.129
	-0.462	6223	66	1.06	1.68	0.131
	0.190	4848	7	0.14	0.47	0.126
	-0.915**	4114	72	1.75	3.05	0.129
ngdom	-0.982**	3804	78	2.05	2.82	0.126
	-0.577**	3364	12	0.36	2.19	0.123
	-1.040**	3155	36	1.14	1.27	0.166
	-1.269**	3098	8	0.26	0.48	0.129
	-0.530*	2310	33	1.43	1.56	0.106
ds	1.902**	1614	7	0.43	0.20	0.133
	-1.748**	1374	7	0.51	0.06	0.143
	0.459	1178	22	1.87	1.64	0.139
	-0.803	1014	122	12.03	10.33	0.169
rakia	-0.177	1012	3	0.30	0.39	0.103
	-0.435	933	4	0.43	3.09	0.094
	0.132	822	16	1.95	1.52	0.145
id	-0.108	808	4	0.50	0.16	0.133
	-0.942**	661	4	0.61	0.10	0.130
	-1.401*	655	19	2.90	1.22	0.091
	-2.954**	654	5	0.76	0.53	0.117
	-2.043	590	1	0.17	1.71	0.155
	-0.612	543	2	0.37	0.00	0.119
a	-1.540	539	1	0.19	0.25	0.141
	-0.387	520	5	0.96	0.23	0.128
	-1.497**	513	26	5.07	16.61	0.135

* "active" mathematician published at least one paper in the period between 1984 and 1989. The active mathematicians are then allocated to the country of their modal "ing" this period. The first-stage coefficient is the coefficient of the interaction between the index of similarity and the post-1992 dummy variable, estimated separately in m . * indicates that the coefficient is between 1.50 and 1.99 times the size of the clustered standard error, and ** indicates that it is at least twice the size of the e . The destination country for the Soviet emigré is the modal country of affiliation between 1992 and 2008. The "immigrant share" is the ratio of the number of ch country to the number of active mathematicians. The "quality-adjusted share" is the ratio of the total number of citations received by the emigrants in 1984-1989 to the number of citations produced by the country's mathematicians in that period. The connectivity index measures the publication link between mathematicians in a particular other countries; see the text for a definition.

(a) Table IX: Soviet diaspora across countries



(b) Figure IX: international heterogeneity

5.13 Table X: Determinants of International Differences in Impact of Soviet Supply Shock

Read:

- Table X regresses each country's first-stage productivity impact on the quality-adjusted immigrant share and connectivity index.
- Both coefficients are negative in the main specifications.
- The R^2 is high for the ten largest countries, indicating that emigre flows and connectivity explain much of the cross-country heterogeneity.

Interpretation. The country-level evidence supports the paper's two-market interpretation: local bodies and global ideas both matter for productivity crowd-out.

Economic message. The same high-skill shock can have different effects across countries depending on migration destinations and integration into global knowledge markets.

TABLE X
DETERMINANTS OF INTERNATIONAL DIFFERENCES IN IMPACT OF SOVIET SUPPLY SHOCK

Sample/variables	Measure of overlap index used in first-stage regression		
	Correlation coefficient	Index of intensity	Index of similarity
Ten largest countries:			
Immigrant share (%)	-0.076 (0.029)	-0.041 (0.022)	-0.382 (0.130)
Connectivity index	-2.431 (0.633)	-1.812 (0.537)	-8.259 (10.188)
R^2	0.806	0.744	0.814
All 26 countries:			
Immigrant share (%)	-0.013 (0.007)	-0.009 (0.005)	-0.062 (0.037)
Connectivity index	-2.609 (0.603)	-1.620 (0.517)	-10.553 (10.672)
R^2	0.230	0.167	0.374

Notes: Robust standard errors in parentheses. The dependent variable is the coefficient of the interaction between the index of overlap and the post-1992 dummy variable from the first-stage regression estimated separately in each country. The regressions in the top panel use the sample of 10 countries where there are at least 2,000 active mathematicians, whereas the regressions in the bottom panel use the 26 countries with at least 500 active mathematicians. The regressions are weighted by the inverse of the sampling variance of the dependent variable. The “immigrant share” is quality-adjusted, giving the ratio of the number of AMS citations received in 1984–1989 by Soviet émigrés who eventually moved to country i to the total number of citations received by mathematicians in country i over the same period. The connectivity index measures the publication link between mathematicians in a particular country and

Table X: determinants of international productivity differences

6 Interpretive synthesis

The paper’s main conceptual point is that knowledge production is not governed only by nonrival idea spillovers. Even when ideas are useful, the institutions surrounding science have scarce resources: faculty slots, journal pages, referees’ attention, seminar slots, and the limited time of frontier researchers. The Soviet influx made new ideas available but also increased competition in those scarce arenas. The empirical evidence suggests that, for highly exposed American mathematicians, the competitive forces dominated on average.

The paper also clarifies why skilled immigration can have complex effects. The incoming Soviet mathematicians were very productive and contributed substantially to U.S. mathematics. But the relevant welfare question is not whether the emigres were productive; it is whether total output rose after accounting for incumbent crowd-out. The aggregate evidence suggests that Soviet output largely filled the gap created by lower incumbent output rather than greatly expanding the total pie.

Finally, the international analysis is important because it rules out a purely local labor-market interpretation. Some countries faced stronger competition in journals because they were more connected to global mathematical publishing. This shows that knowledge producers compete in overlapping markets: a geographic labor market and a global market for codified ideas.

7 Short summary

- Borjas and Doran study how the collapse of the Soviet Union changed global mathematics.

- The paper uses MathSciNet, AMS, ISI, and Mathematics Genealogy Project data to track individual mathematicians, publications, citations, fields, affiliations, and students.
- The key treatment is pre-1990 overlap between an American mathematician’s fields and Soviet pre-collapse fields.
- Soviet and American mathematics were highly specialized in different fields before the collapse.
- Soviet-American coauthorship and contact rose sharply after 1990.
- Soviet emigres were positively selected and highly productive.
- The main regressions show that high-overlap American mathematicians published fewer papers and received fewer citations after 1992.
- The negative effect survives alternative exposure indices, alternative samples, short- and long-run windows, and field-year fixed effects.
- Highly exposed American mathematicians were less likely to produce home-run papers.
- They were more likely to stop publishing and to move to lower quality institutions.
- Students of Soviet emigres benefited, but students of exposed American advisors did worse.
- Aggregate Soviet output largely filled in the decline among incumbent Americans, with little evidence of a major expansion of total output.
- Cross-country evidence shows that both immigrant share and journal-market connectivity explain the size of the productivity effect.
- The central lesson is that high-skilled inflows can create both spillovers and crowd-out, and the net effect depends on resource scarcity in knowledge markets.

8 Conclusion

8.1 Core conclusion

The collapse of the Soviet Union created a large, unexpected inflow of highly skilled Soviet mathematicians into the United States and the world. American mathematicians whose pre-1990 research most overlapped with Soviet research experienced lower productivity after the shock. The evidence appears in papers, citations, home-run papers, publishing survival, institutional mobility, and aggregate output accounting.

8.2 Economic intuition

The intuitive mechanism is a tension between spillovers and scarcity. Soviet mathematicians brought new ideas, methods, and training. But they also competed for a limited set of jobs, journals, citations, and intellectual attention. In the exposed fields, those scarce resources appear to have bound strongly enough that the net effect on incumbents was negative.

8.3 Incremental takeaway

The paper’s incremental contribution is to show empirically, in a rich individual-level data set, that the arrival of world-class knowledge producers can reduce the productivity of closely competing incumbents. This does not contradict the existence of knowledge spillovers. Instead, it shows that the production of knowledge is embedded in markets and institutions where crowd-out can be quantitatively important.

8.4 What the paper does not claim

The paper does not claim that Soviet mathematicians were harmful to mathematics or that ideas are rival in the usual sense. It also does not claim that all high-skilled immigration has negative effects. The evidence is specific to a context with a large and sudden inflow, highly specialized research overlap, and scarce publication and academic employment opportunities.

8.5 Final takeaway

Knowledge markets are not frictionless idea machines. They are human institutions with limited attention, limited jobs, and limited channels of recognition. The paper's lasting lesson is that the effect of high-skilled immigration on incumbent knowledge workers depends on both the productive value of new ideas and the scarcity of the resources needed to turn ideas into recognized output.